MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)

THIRD SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG3DSEBTS201- AQUARIUM MANAGEMENT

Duration: 2hrs Maximum Marks: 70

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

Part A
Fill in the blanks
Answer All Questions
Each question carries 1 mark

1. Marine ornamental fish are often collected from _____ and _____ ecosystems.**[U][CO1]**
2. The diversity of ornamental fish includes species from both _____ and _____ ecosystems.**[U][CO1]**
3. _____ is a major exporting state of ornamental fish in India.**[U][CO1]**
4. The word aquarium is derived from the Latin words aqua meaning _____ and rium meaning _____.**[K][CO1][CO2]**
5. A _____ is a hybrid setup combining both aquatic and terrestrial elements, often used to simulate wetlands.**[U][CO1][CO2]**
6. The role of _____ is to educate the public about aquatic life and promote conservation.**[U][CO1][CO2]**
7. _____ are used in aquariums to supply oxygen by creating bubbles. **[K][CO3]**
8. _____ are used in aquaria to maintain low temperatures in tropical or marine setups.**[K][CO3]**
9. The _____ is a specialized structure used by male livebearers for reproduction.**[K][CO3]**
10. Popular carp species like Rohu, catla and mrigal are the examples of ---------- **[K][CO3]**

[10 x 1 = 10]

Part B

Short questions Type Questions

Answer 10 Questions

Each question carries3 marks

11. Give any three characteristics of marine ornamental fishes.**[U] [CO1]**
12. Briefly explain the concept of a reef aquarium. **[K] [CO1][CO2]**
13. Discuss the diversity of ornamental fish in India.**[U][CO1]**
14. List three functions of public aquaria.**[K][CO1][CO2]**
15. Why is filtration important in aquarium maintenance?**[U][CO3]**

16. Differentiate between mechanical and biological filters.**[U][CO3]**
17. What is the importance of live feed culture?cite any one example of live food source used in aquaculture**[K][CO2][CO3]**
18. What is the significance and challenges of maintain ornamental aquatic plants**? [U][CO3]**
19. Briefly describe larval rearing in ornamental fish.**[U][CO3]**
20. What are the different methods to maintain an effective health management in aquarium fishes? **[K][CO2] [CO3]**
21. What are the different nutrient and environmental requirements required for the growth of ornamental aquatic plants**[U][CO3]**
22. Cite any two examples of common diseases affecting both fresh water and marine ornamental fishes **[K][CO2] [CO3]**

[10x 3= 30]

Part C

Short Essay Type Questions

Answer any six questions
Each question carries 5 marks

23. Describe the major ornamental fish species of India with their common features.**[U][CO3]**
24. Explain the principles involved in setting up and maintaining a healthy aquarium. **[K][CO3].**
25. Explain the latest trends in aquarium keeping.**[U][CO3]**
26. What are the challenges in ornamental fish seed production?**[U][CO3]**
27. Explain the process of sex determination and sexual dimorphism in ornamental fishes.
28. What are the different steps in preparing and maintain a fish aquarium**[K][CO2] [CO3]**
29. What are the significances of formulated feed and their advantages**[K][CO2] [CO3]**
30. Describe the key features to be considered in the construction of a breeding tank for ornamental fishes.[K][CO3]**[6x5=30]**

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